

Markov Chains

Handwritten Lecture Notes

ASM Course

Week Beyond Midterm

Topics: Stationary Distributions · Detailed Balance · Irreducibility · Aperiodicity · Ergodicity ·
Convergence

Problem Statement

A distn Π (target).

Want to generate r.v.'s from Π :

$$X \sim \Pi, \quad \mu = \mathbb{E}[X].$$

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \Pi \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \xrightarrow{\text{a.s.}} \mu \text{ (by SLLN)}$$

Cannot generate from Π either directly or indirectly.

- Find a Markov chain with transition matrix P which satisfies $\Pi P = \Pi$
- Generate a Markov chain $Y_1, Y_2, \dots, Y_n$ from P
- Under certain conditions,

$$\frac{1}{n} \sum_{j=1}^n Y_j \rightarrow \mu$$

$x, y \rightarrow$ states of Markov chain

$$P = ((p_{x,y}))$$

$$p_{x,y} = \left(\frac{w(y)}{w(x)} \wedge 1 \right) / d$$

Do not know Π . But know it upto a const i.e.

$$\Pi(x) = c w(x)$$

$d \geq \max$ number of states that you can go to in 1 step from any x

$$x \longrightarrow \{\text{multiple states}\}$$

say dim,

$$\boxed{n \times k}$$

$$d = m(m-1)k(k-1)$$

States of a Markov Chain

$$P = ((p_{x,y})) \quad \text{and} \quad p_{x,y} = \left(\frac{w(y)}{w(x)} \wedge 1 \right) / d.$$

Do not know Π . But know it up to a constant i.e.

$$\Pi(x) = c w(x).$$

1.

$d \geq \max\{\text{number of states that you can go to in 1 step from any } x\}.$

2. Restrict to $\{x : \Pi(x) > 0\}.$

(3) What if the

$$\begin{bmatrix} (i,j)^+ & (i,j)^- \\ (i',j')^- & (i',j')^+ \end{bmatrix}$$

(i,j) -th element of X is zero?

Then stay at x .

Choose i, i', j, j' again from $n \times k$ table.

$$x \rightarrow y$$

Proposition: The transition probability matrix

$$p(x, y) = \begin{cases} \left(\frac{w(y)}{w(x)} \wedge 1 \right) / d, & y \neq x, y \in \mathcal{T} \\ 1 - \sum_{y \neq x} p(x, y), & y = x \end{cases}$$

is aperiodic and irreducible on $\mathcal{T}$ and its unique stationary

Stationarity and Detailed Balance

As aperiodic and irreducible on Π and its unique stationary distribution is Π (from the eg).

Pf: Π is a stationary distn of P .

We have shown this in 2 steps:

1. P satisfies the detailed balance eqn:

$$\Pi(x)p(x, y) = \Pi(y)p(y, x), \quad \forall x, y.$$

2. If P satisfies detailed balance eqn in Π then it satisfies

$$\Pi P = \Pi.$$

For this eg: Aperiodic. There's some $x \in \mathcal{T}$ s.t. $p(x, x) > 0$.

Pick x that has one element 0.

If (i, j') combination is that element then algorithm dictates "stay at x ". So $p(x, x) > 0$.

Irreducible: You can reach any state y from any other state x . (Tutorial)

General result: Aperiodic & irreducible $P \Rightarrow \Pi$ is unique.

Thm: Let $\{Y_n\}$ be aperiodic, irreducible, positive recurrent Markov chain on a finite or countable state space $\mathcal{X}$ with stationary distn Π and let f be a real valued fn on $\mathcal{X}$ such that

$$\sum_{x \in \mathcal{X}} |f(x)| \Pi(x) < \infty,$$

then for any initial Markov chain on a finite or countable state space $\mathcal{X}$, stationary distn Π and let f be a real valued fn such that

$$\sum_{x \in \mathcal{X}} |f(x)| \Pi(x) < \infty,$$

then for any initial state x , with P^x probability one,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n f(Y_j) = \mu := \sum_{x \in \mathcal{X}} f(x) \Pi(x).$$

where $Y_1, Y_2, \dots$ is a sequence from P .

$$X_n, \dots, X_\pi \sim \Pi$$

$$\frac{1}{n} \sum_{i=1}^n f(X_i) \rightarrow \mu$$

By SLLN

Requires Ergodic Thm.

Modifications: d was crucial.

For a general state space $\mathcal{X}$, d may be infinite.

Instead of this $p(x, y) \propto \frac{w(y)}{w(x)}$

we want a diff P that leads to faster convergence.

$$p(x, y) = \left(\frac{\Pi_y}{\Pi_x} \cdot \frac{q(y, x)}{q(x, y)} \wedge 1 \right) q(x, y)$$

where $Q = ((q_{x,y}))$ is some transition matrix.

$$p(x, y) = \begin{cases} \left(\frac{\Pi_y}{\Pi_x} \cdot \frac{q(y, x)}{q(x, y)} \wedge 1 \right) q(x, y), & y \neq x \\ 1 - \sum_{y \neq x} p(x, y), & y = x \end{cases}$$

where $Q = ((q_{x,y}))$ is some transition matrix.

1. $((p_{x,y})) = P$ is a transition matrix
2. P satisfies detailed balance eqn.
3. $\Pi P = \Pi$

Previous definition was special case of the one with $q(x,y)=1/d$

Continuous case: $\mathcal{X}$ is uncountable and Π is density on $\mathcal{X}$.

Let Q be a transition fn s.t. for every $x \in \mathcal{X}$, Q has a density $q(x, \cdot)$, define transition density as

$$p(x, y) = \left(\frac{\Pi(y)}{\Pi(x)} \frac{q(y, x)}{q(x, y)} \wedge 1 \right) q(x, y)$$

Choice of q : In Metropolis (1953), Q was chosen to be a symmetric matrix/density. Eg for density Q could be normal

$$q(x, y) = q(y, x)$$

$$Y \sim N(x, 1) \quad X \sim N(y, 1)$$

$$\frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2}(y-x)^2 \right\} \quad \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2}(x-y)^2 \right\}$$

normal

$$q(x, y) = q(y, x)$$

$$Y \sim N(x, 1) \quad X \sim N(y, 1)$$

$$\frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2}(y-x)^2 \right\} = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2}(x-y)^2 \right\}$$

Some papers choose

$$q(x, y) = q(|x - y|)$$

Normal also satisfies this.

Scale is important. Determines how quickly the chain moves / mixes / covers the whole space.

Very large scale may lead to convergence while a small scale still move in the same neighborhood.